

Mapa

GeoGuessr on Stellar Soroban

Guess locations from street view. Win XLM.

Live: mapa-gamma-seven.vercel.app

The Problem

- Centralized game servers can be shut down or manipulated
- Prize pools controlled by a single entity
- No transparency in scoring or payouts
- Users don't truly own in-game assets

The Solution

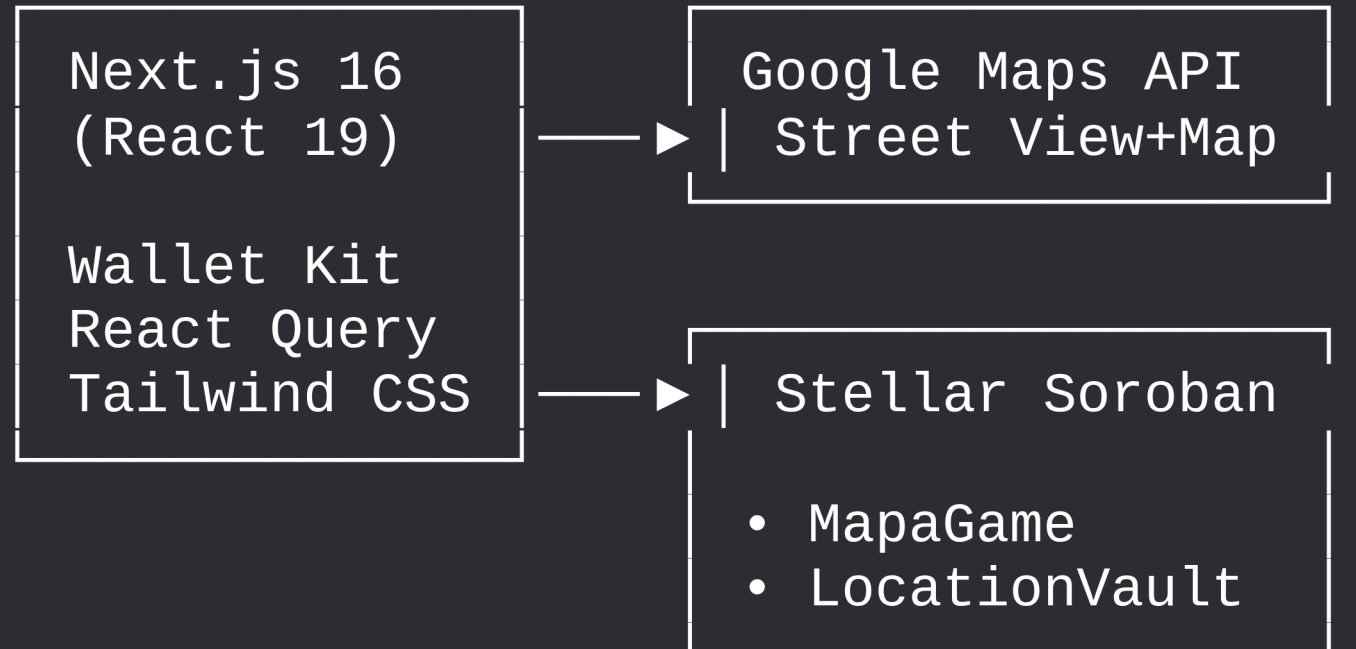
Decentralized geography guessing on Stellar Soroban

- Smart contract-governed game logic
- Transparent, on-chain prize distribution
- Wallet-based identity — no signups
- Every guess, stake, and payout verifiable on ledger

How It Works

1. **Connect Wallet** — Freightier, xBull, Lobstr, or Hana
2. **See a Location** — Random Google Street View imagery
3. **Drop a Pin** — Guess where in the world it is
4. **Win XLM** — Closer guess = bigger payout from the pot

Architecture



Smart Contracts

MapaGame

- Room lifecycle: create, join, guess, resolve, claim
- Haversine distance with integer math
- Prize: 250bps platform fee, rest to winner

MapaLocationVault

- Location storage (lat/lng + images)
- Random location selection
- Admin CRUD

Contract Addresses

Contract	Address
MapaGame	CCBTZEMT...
LocationVault	CC4RZMXH...
MAPATST2 Token	CD35R3Y5...

Deployed on **Stellar Testnet**

Prize Mechanics

- **Entry fee:** 0.1 MAPATST2 per game
- **Score:** 0–1,000,000 (perfect = 1M, >20km = 0)
- **Prize:** $\text{entry_fee} \times \text{score} / 1,000,000$
- **Tie:** Equal distance → both refunded

Tech Stack

Layer	Technology
Blockchain	Stellar Soroban (Rust)
Frontend	Next.js 16, React 19, TS 5
Styling	Tailwind CSS v4
Maps	Google Maps API (Street View)
Wallet	Stellar Wallet Kit 2
Data	TanStack React Query 5
CI/CD	GitHub Actions + Vercel

Testing

- 11 unit tests via `cargo test`
- Room lifecycle (create → join → guess → resolve → claim)
- Auto-match with fee calculation
- Tie resolution
- GitHub Actions CI

User Feedback

50 testnet wallet holders submitted feedback via Google Form.

Top requests:

1. QR code integration
2. Onboarding improvements
3. Search / filter rooms
4. Leaderboards
5. Friend challenges

Future Roadmap

- Mainnet deployment
- QR code for wallet sharing
- Onboarding tutorial
- Leaderboards & rankings
- Group/friend challenges
- Mobile app (React Native)

Get Involved

- **GitHub:** github.com/rylsherdamz-rgb/mapa
- **Feedback:** Google Form
- **License:** MIT

Built on Stellar Soroban